

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### Product Identifier

Perihalan Produk: **Poly(ethylene oxide), approximate M.W. 600.000**  
Product Description: **Poly(ethylene oxide), approximate M.W. 600.000**  
Cat No. : 178610000; 178610050; 178612500; 178615000  
Synonyms PEO  
CAS No 25322-68-3

### Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.  
Uses advised against No Information available

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## SECTION 2: HAZARDS IDENTIFICATION

### Classification of the substance or mixture

### Label Elements

#### Hazard Statements

#### Response

P332 + P313 - If skin irritation occurs: Get medical advice/attention  
P337 + P313 - If eye irritation persists: Get medical advice/attention  
P362 + P364 - Take off contaminated clothing and wash it before reuse

### Other Hazards

This product does not contain any known or suspected endocrine disruptors

# SAFETY DATA SHEET

Poly(ethylene oxide), approximate M.W. 600.000

Revision Date 21-Mar-2025

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component           | CAS No     | Weight % |
|---------------------|------------|----------|
| Polyethylene glycol | 25322-68-3 | 100      |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

**Eye Contact** Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact** Wash off immediately with plenty of water for at least 15 minutes. Get medical attention.

**Ingestion** Do NOT induce vomiting. Get medical attention.

**Inhalation** Remove to fresh air. Get medical attention. If not breathing, give artificial respiration.

**Self-Protection of the First Aider** No special precautions required.

### Most important symptoms and effects, both acute and delayed

No information available.

### Indication of any immediate medical attention and special treatment needed

**Notes to Physician** Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### **Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

#### **Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

# SAFETY DATA SHEET

Poly(ethylene oxide), approximate M.W. 600.000

Revision Date 21-Mar-2025

Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation.

## Environmental precautions

Avoid release to the environment. See Section 12 for additional Ecological Information.

## Methods and Material for Containment and Cleaning Up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

## Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation. Avoid dust formation.

### Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component           | European Union | The United Kingdom | Germany                                                                                                                                                                                                                                                                                                                     |
|---------------------|----------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Polyethylene glycol |                |                    | TWA: 200 mg/m <sup>3</sup> (8 Stunden). AGW<br>- exposure factor 2<br>TWA: 250 mg/m <sup>3</sup> (8 Stunden). MAK<br>average molecular weight<br>200-600; because formation of a<br>mist is possible, exposure should be<br>minimized for reasons of<br>occupational safety and hygiene<br>Höhepunkt: 500 mg/m <sup>3</sup> |

### Exposure Controls

#### Engineering Measures

None under normal use conditions.

### Personal protective equipment

#### **Eye Protection**

Wear safety glasses with side shields (or goggles)

#### **Hand Protection**

Protective gloves

#### **Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.  
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g.

# SAFETY DATA SHEET

Poly(ethylene oxide), approximate M.W. 600.000

Revision Date 21-Mar-2025

sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

## Respiratory Protection

### Recommended Filter type:

No protective equipment is needed under normal use conditions

Particle filter

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

## Environmental exposure controls

No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

#### Appearance

White

#### Physical State

Powder Solid

#### Odor

Slight

#### Odor Threshold

No data available

#### pH

No information available

#### Melting Point/Range

No data available

#### Softening Point

No data available

#### Boiling Point/Range

No information available

#### Flash Point

No information available

Method - No information available

#### Evaporation Rate

Not applicable

Solid

#### Flammability (solid,gas)

No information available

#### Explosion Limits

No data available

#### Vapor Pressure

No information available

#### Vapor Density

Not applicable

Solid

#### Specific Gravity / Density

No data available

#### Bulk Density

No data available

#### Water Solubility

Soluble

#### Solubility in other solvents

No information available

#### Partition Coefficient (n-octanol/water)

#### Autoignition Temperature

Not applicable

#### Decomposition Temperature

No data available

#### Viscosity

Not applicable

Solid

#### Explosive Properties

No information available

#### Oxidizing Properties

No information available

## SECTION 10: STABILITY AND REACTIVITY

# SAFETY DATA SHEET

Poly(ethylene oxide), approximate M.W. 600.000

Revision Date 21-Mar-2025

## Reactivity

None known, based on information available.

## Chemical Stability

Stable under normal conditions.

## Possibility of Hazardous Reactions

### **Hazardous Polymerization Hazardous Reactions**

Hazardous polymerization does not occur.  
None under normal processing.

## Conditions to Avoid

Avoid dust formation. Incompatible products. Excess heat.

## Incompatible Materials

Strong oxidizing agents.

## Hazardous Decomposition Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### **Product Information**

#### **(a) acute toxicity;**

**Oral**

Based on available data, the classification criteria are not met

**Dermal**

Based on available data, the classification criteria are not met

**Inhalation**

Based on available data, the classification criteria are not met

### Toxicology data for the components

| Component           | LD50 Oral              | LD50 Dermal               | LC50 Inhalation |
|---------------------|------------------------|---------------------------|-----------------|
| Polyethylene glycol | LD50 = 22 g/kg ( Rat ) | LD50 > 20 g/kg ( Rabbit ) | -               |

#### **(b) skin corrosion/irritation;**

No data available

#### **(c) serious eye damage/irritation;**

No data available

#### **(d) respiratory or skin sensitization;**

**Respiratory**

No data available

**Skin**

No data available

| Component                                 | Test method                                            | Test species | Study result    |
|-------------------------------------------|--------------------------------------------------------|--------------|-----------------|
| Polyethylene glycol<br>25322-68-3 ( 100 ) | in vivo: Test method Human<br>Repeat Insult Patch Test | Man          | non-sensitising |

# SAFETY DATA SHEET

Poly(ethylene oxide), approximate M.W. 600.000

Revision Date 21-Mar-2025

(e) germ cell mutagenicity; No data available

| Component                                 | Test method             | Test species | Study result |
|-------------------------------------------|-------------------------|--------------|--------------|
| Polyethylene glycol<br>25322-68-3 ( 100 ) | OECD Test Guideline 471 | in vivo      | negative     |

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; Not applicable  
Solid

Other Adverse Effects The toxicological properties have not been fully investigated.

Symptoms / effects, both acute and delayed No information available.

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity effects Do not empty into drains. .

| Component           | Freshwater Fish                                                     | Water Flea                                                    | Freshwater Algae                                                           | Microtox |
|---------------------|---------------------------------------------------------------------|---------------------------------------------------------------|----------------------------------------------------------------------------|----------|
| Polyethylene glycol | LC50 > 100 mg/L 96h,<br>(Poecilia reticulata)<br>OECD Guideline 203 | EC50 > 100 mg/L 48h,<br>(Daphnia magna)<br>OECD Guideline 202 | EC50 > 100 mg/L 96h,<br>(Scenedesmus<br>subspicatus) OECD<br>Guideline 201 |          |

Persistence and degradability  
Persistence Soluble in water, Persistence is unlikely, based on information available.

Bioaccumulative potential Bioaccumulation is unlikely

Mobility in soil The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors

Other adverse effects No information available

# SAFETY DATA SHEET

Poly(ethylene oxide), approximate M.W. 600.000

Revision Date 21-Mar-2025

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

#### **Waste from Residues/Unused Products**

Dispose of in accordance with local regulations

#### **Contaminated Packaging**

Empty containers should be taken to an approved waste handling site for recycling or disposal

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

Not regulated

### Road and Rail Transport

Not regulated

### IATA

Not regulated

### **Special Precautions for User**

No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### **International Inventories**

X = listed

| Component           | EINECS | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|---------------------|--------|------|-----|-------|------|------|-------|------|----------|
| Polyethylene glycol | -      | X    | X   | X     | X    | X    | X     | X    | KE-20228 |

### National Regulations

#### **Persistent Organic Pollutant**

This product does not contain any known or suspected substance

#### **Ozone Depletion Potential**

This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

# SAFETY DATA SHEET

Poly(ethylene oxide), approximate M.W. 600.000

Revision Date 21-Mar-2025

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Revision Date

21-Mar-2025

Revision Summary

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**